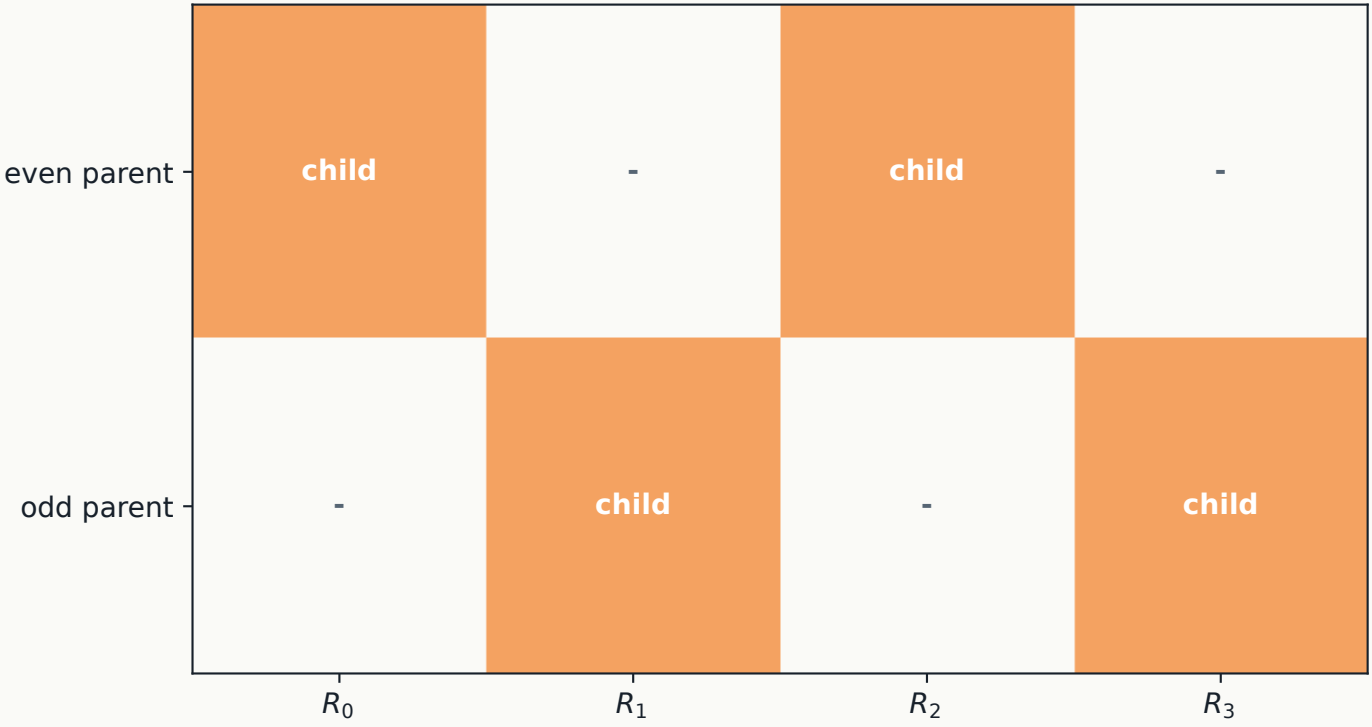
The coarse-to-fine incidence matrix is exact, not statistical



Coarsening sums the indicated child packets; refinement expands each parent into exactly those descendants.